A Level • Cambridge (CIE) • Chemistry

24 mins 13 questions

Multiple Choice Practice

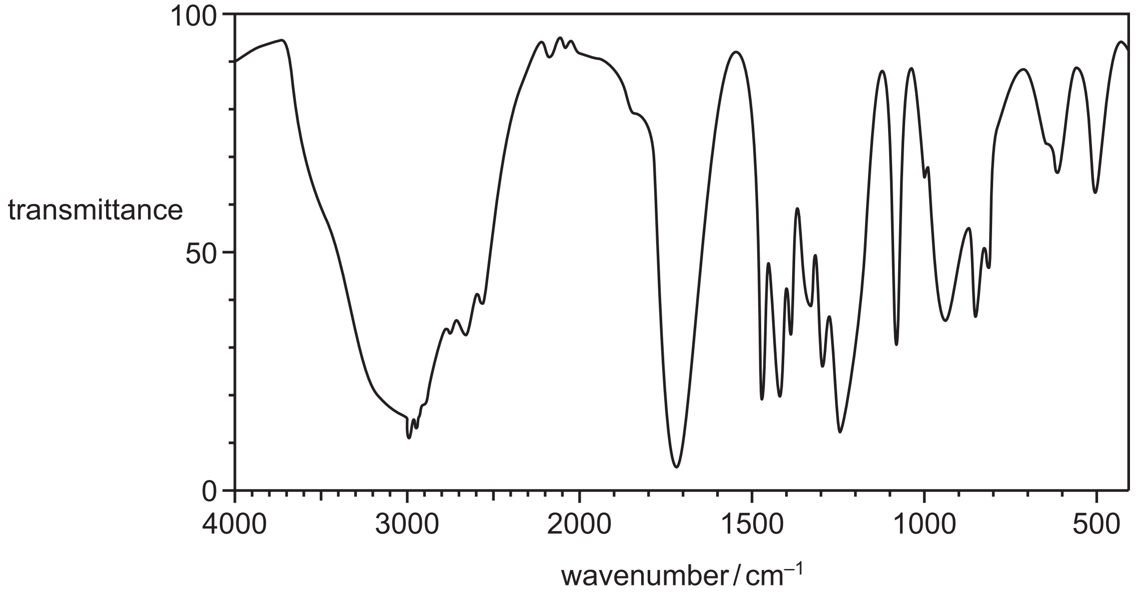
Topical Practice MCQ

22. Analytical techniques

Carefully selected past paper questions by topic.

40 Compound X reacts with acidified $K_2Cr_2O_7$ to form compound Y.

The infrared spectrum of compound Y is shown.



bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

What is the identity of compound X?

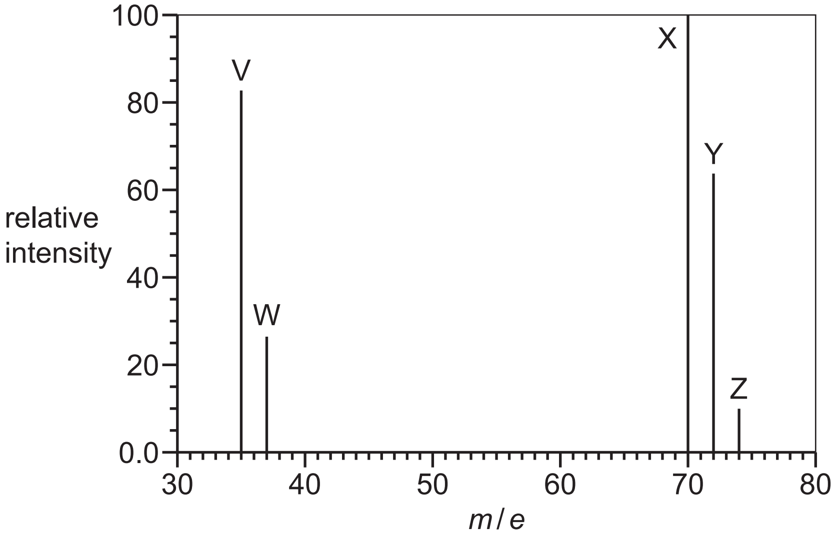
A propan-1-ol

B propan-2-ol

C propanone

D propanoic acid

40 The diagram shows the mass spectrum of a sample of chlorine. Peaks V, W, X, Y and Z are labelled.



Which statements about this spectrum are correct?

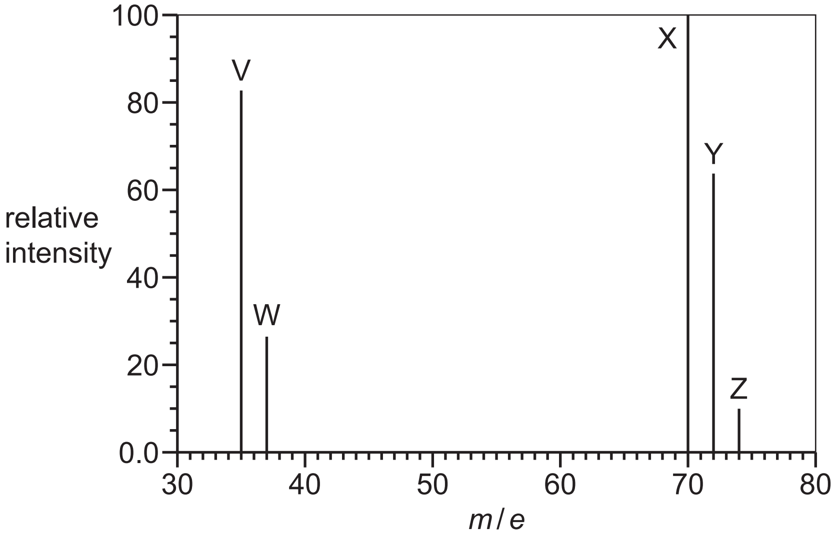
1 The relative atomic mass of chlorine can be calculated from the abundances and *m* / *e* values of 2 of the 5 peaks.

2 37.0 g of the species responsible for peak Z contains 3.011×10^{23} molecules.

3 The relative molecular mass of chlorine can be calculated from the abundances and *m* / *e* values of peaks X, Y and Z.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

40 The diagram shows the mass spectrum of a sample of chlorine. Peaks V, W, X, Y and Z are labelled.



Which statements about this spectrum are correct?

1 The relative atomic mass of chlorine can be calculated from the abundances and *m* / *e* values of 2 of the 5 peaks.

2 37.0 g of the species responsible for peak Z contains 3.011×10^{23} molecules.

3 The relative molecular mass of chlorine can be calculated from the abundances and *m* / *e* values of peaks X, Y and Z.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

40 The mass spectrum of compound X has M, M+1 and M+2 peaks. Other peaks are also present.

Peak M is the molecular ion peak, M^+. Peak M has a relative abundance fifteen times that of peak M+1.

Peaks M and M+2 are of equal height.

What could be compound X?

A 1-chloro-2,2-dimethylpentane

B 2-chloro-3-methylpentane

C 2-bromo-2-methylhexane

D 3-bromo-2,2-dimethylbutane

40 Oxygen has three stable isotopes, ^{16}O, ^{17}O and ^{18}O. All three isotopes are present in a sample of oxygen gas, O_2, which was analysed using a mass spectrometer.

How many peaks associated with the O_2^+ ion would be expected?

A 3 **B** 5 **C** 6 **D** 9

40 Three organic compounds are listed.

1 ethanal

2 propan-1-ol

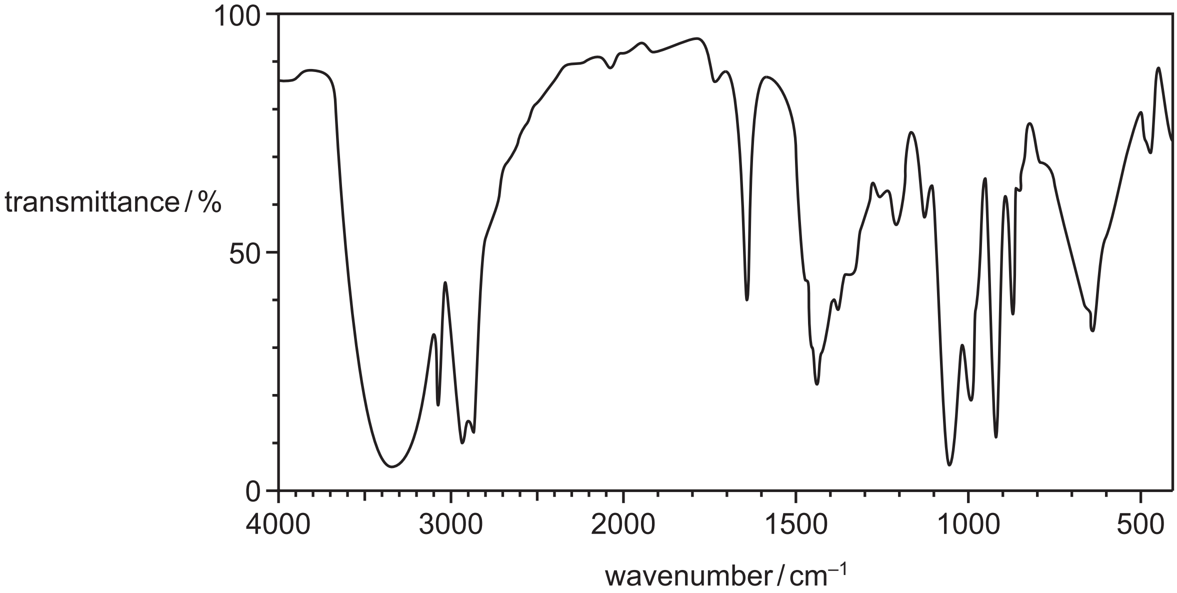
3 propan-2-ol

Which compounds will have a mass spectrum that contains a fragment peak at m/e = 43?

A 1 only **B** 1 and 2 only **C** 2 and 3 only **D** 1, 2 and 3

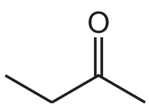
30 The molecular formula of Z is C_4H_8O.

The infra-red spectrum of Z is shown.

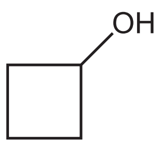


What could be Z?

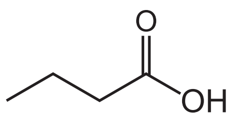
A



B



C



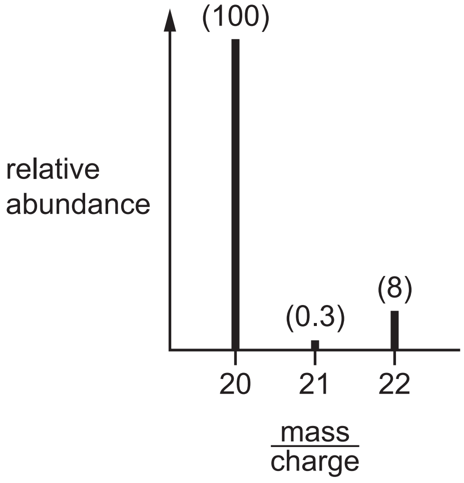
D

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against

1 The mass spectrum of a sample of neon is shown. The relative abundance of each peak is written in brackets above it.



What is the relative atomic mass, A_r, of this sample of neon?

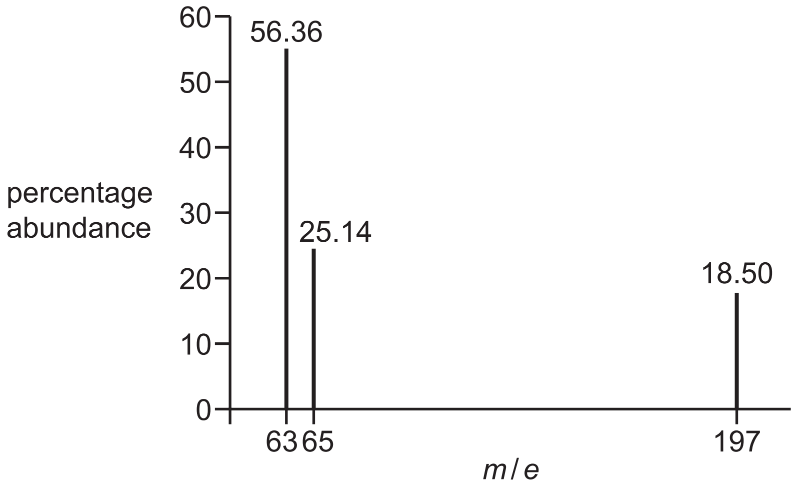
A 20.15 **B** 20.20 **C** 21.00 **D** 21.82

2 Oxygen has three stable isotopes, ^{16}O, ^{17}O and ^{18}O. All three isotopes are present in a sample of oxygen gas, O_2, which was analysed using a mass spectrometer.

How many peaks associated with the O_2^+ ion would be expected?

A 3 **B** 5 **C** 6 **D** 9

2 The mass spectrum of an alloy of copper and gold is shown.



Which expression can be used to calculate the relative atomic mass, A_r, of copper present in this sample?

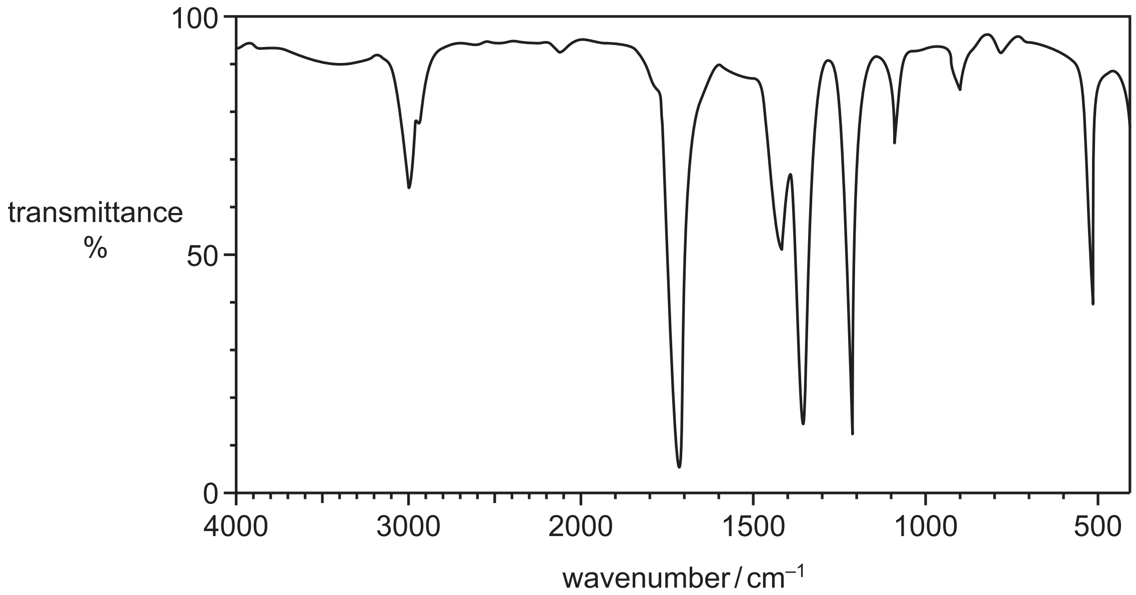
A $\dfrac{(56.36 \times 63) + (25.14 \times 65)}{(56.36 + 25.14 + 18.50)}$

B $\dfrac{(56.36 \times 63) + (25.14 \times 65) + (18.50 \times 197)}{(56.36 + 25.14 + 18.50)}$

C $\dfrac{(56.36 \times 63) + (25.14 \times 65)}{(56.36 + 25.14)}$

D $\dfrac{(56.36 \times 63) + (25.14 \times 65)}{(63 + 65)}$

30 The infra-red spectrum of an organic compound is shown.



Which compound could give this spectrum?

A $CH_3CH_2CO_2H$

B $CH_3CH(OH)CH_3$

C CH_3COCH_3

D CH_3COCH_2OH

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

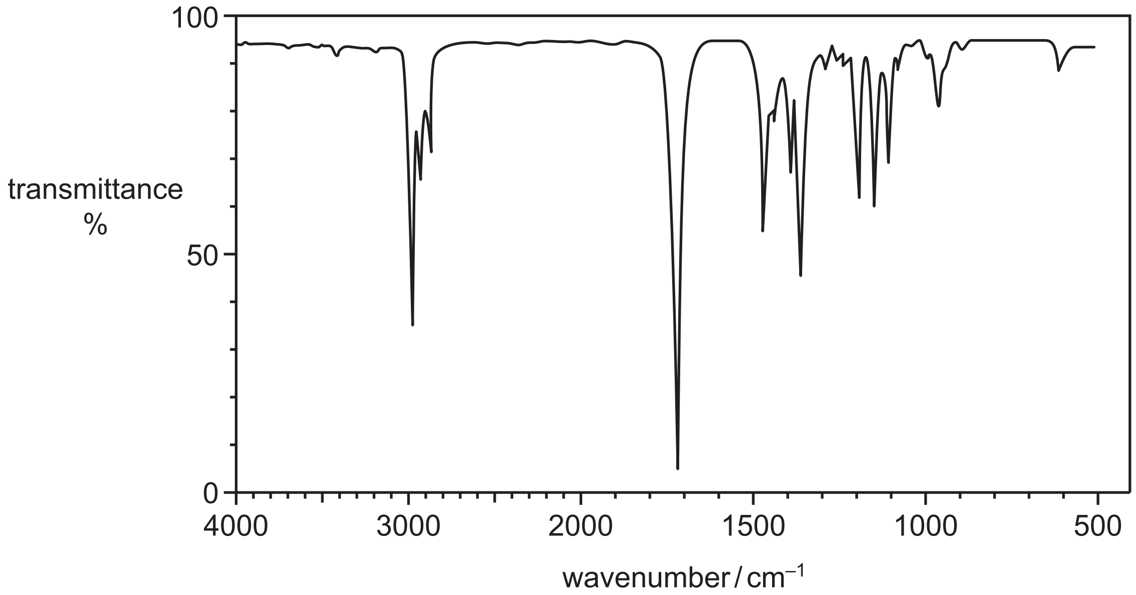
A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

Use of the Data Booklet may be appropriate for some questions.

30 **J** is a branched-chain alcohol, $C_5H_{12}O$. **J** is heated under reflux with an excess of $Cr_2O_7^{2-}/H^+$ until no further reaction occurs. An organic compound **K** is formed in good yield.

The infra-red spectrum of **K** is shown.



What are the structures of the branched-chain alcohol **J** and compound **K**?

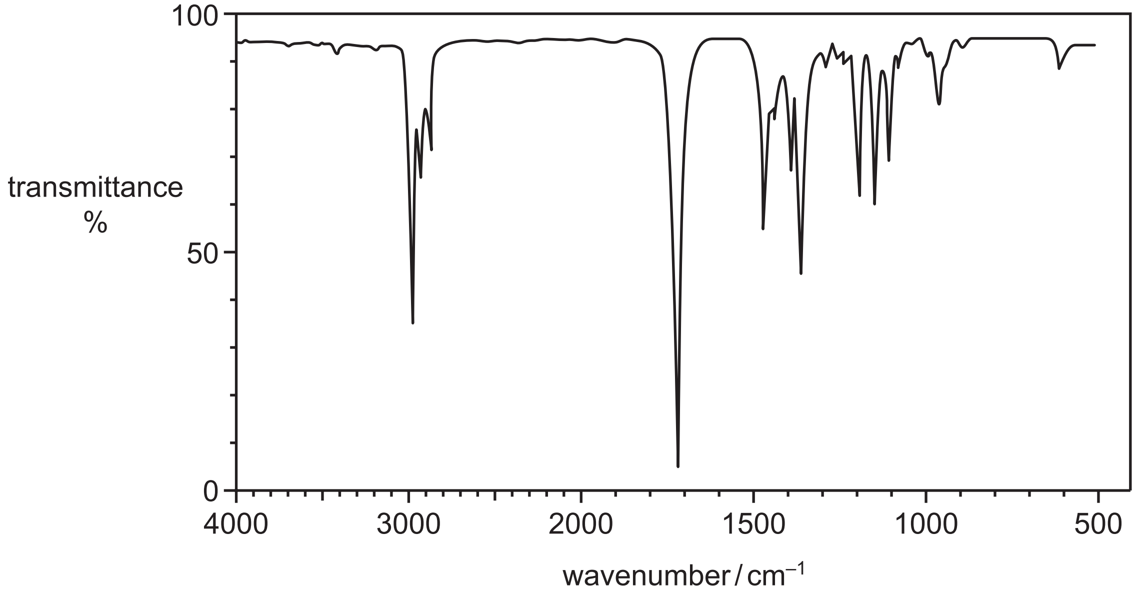
	J	**K**
A	$CH_3CH(CH_3)CH_2CH_2OH$	$CH_3CH(CH_3)CH_2CHO$
B	$CH_3CH_2CH(OH)CH_2CH_3$	$CH_3CH_2COCH_2CH_3$
C	$CH_3CH(CH_3)CH(OH)CH_3$	$CH_3CH(CH_3)COCH_3$
D	$CH_3CH(CH_3)CH_2CH_2OH$	$CH_3CH(CH_3)CH_2COOH$

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

30 **J** is a branched-chain alcohol, $C_5H_{12}O$. **J** is heated under reflux with an excess of $Cr_2O_7^{2-}/H^+$ until no further reaction occurs. An organic compound **K** is formed in good yield.

The infra-red spectrum of **K** is shown.



What are the structures of the branched-chain alcohol **J** and compound **K**?

	J	**K**
A	$CH_3CH(CH_3)CH_2CH_2OH$	$CH_3CH(CH_3)CH_2CHO$
B	$CH_3CH_2CH(OH)CH_2CH_3$	$CH_3CH_2COCH_2CH_3$
C	$CH_3CH(CH_3)CH(OH)CH_3$	$CH_3CH(CH_3)COCH_3$
D	$CH_3CH(CH_3)CH_2CH_2OH$	$CH_3CH(CH_3)CH_2COOH$

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Answer Key

No.	Paper	Topic	Answer
1	9701_s24_qp_12	22. Analytical techniques	**A**
2	9701_w24_qp_11	22. Analytical techniques	**A**
3	9701_w24_qp_13	22. Analytical techniques	**A**
4	9701_s23_qp_13	22. Analytical techniques	**D**
5	9701_w23_qp_12	22. Analytical techniques	**B**
6	9701_s22_qp_12	22. Analytical techniques	**D**
7	9701_s21_qp_12	22. Analytical techniques	**D**
8	9701_w21_qp_11	22. Analytical techniques	**A**
9	9701_s19_qp_12	22. Analytical techniques	**B**
10	9701_s19_qp_13	22. Analytical techniques	**C**
11	9701_s18_qp_12	22. Analytical techniques	**C**
12	9701_w18_qp_11	22. Analytical techniques	**C**
13	9701_w18_qp_13	22. Analytical techniques	**C**